

Ritvik Aggarwal

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EDUCATION

University of Toronto

Honours Bachelor of Science in Computer Science & Mathematics

Toronto, ON, Canada

September. 2025 – September 2029

EXPERIENCE

Machine Learning Researcher

Algoverse AI Research

May 2026 – Present

Remote

- Currently researching trustworthy agentic AI and synthetic data generation, with a focus on Agentic Self-Instruct systems. Exploring how adversarial challengers can exploit recurring verifier failure modes, and how adaptive verification methods can improve robustness over time.

Machine Learning Researcher - Co-lead

Algoverse AI Research

July 2024 – Nov 2024

Remote

Accepted to NAACL SRW '25, SoCal NLP Symposium '24

- Developed improvements to retrieval-augmented generation (RAG) systems, enhancing the accuracy of information retrieval and response generation.
- Designed a chunk-level filtering mechanism that discards irrelevant content at a granular level, outperforming traditional document-level retrieval baselines.
- Introduced novel techniques including relevance scoring, query rewriting, and redundancy removal to further boost system performance.
- Evaluated on PopQA, PubHealth, and Biography benchmarks; achieved 64.9% accuracy on PopQA, outperforming the nearest baseline (CRAG) by 10 percentage points.
- Demonstrated scalability and domain-agnostic performance, with particular gains on precision-sensitive tasks such as fact-checking and multi-hop reasoning.

Vice-President Engineering

University of Toronto Algorithmic Trading Society

May. 2026 – Present

Toronto, ON, Canada

- Lead the engineering function of a student society dedicated to educating members in quantitative finance, algorithmic trading, market microstructure, and financial systems engineering.
- Oversee development and maintenance of technical infrastructure including code repositories, software systems, and engineering standards used across trading projects and workshops.
- Organize and coordinate technical workshops covering quantitative finance and systems engineering topics, enabling hands-on simulated learning experiences for members.
- Build and manage cross-functional technical project teams, guiding students through collaborative, competition-ready algorithmic trading systems.

PROJECTS

Predictive Analysis of Housing Prices Using Linear Regression and Decision Tree Models

Python, Pandas, NumPy, Matplotlib, Seaborn

Sep 2024 – Mar 2025

- Performed comprehensive data preprocessing including identification of numerical, categorical, and binary columns, and dataset optimization to handle data imbalance through hyperparameter tuning and K-fold cross-validation
- Performed comprehensive data preprocessing including identification of numerical, categorical, and binary columns, and dataset optimization to handle data imbalance
- Evaluated model performance using regression metrics (**MAE**, **MSE**, **RMSE**, **R²**) and classification metrics (Precision, Recall, F1), with early stopping and patience mechanisms
- Applied **bias-variance assessment** and feature correlation analysis using heat maps to improve model generalization & implemented **SHAP** and **LIME** techniques for model interpretability and correlation analysis, identifying key housing price predictors

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, JavaScript, HTML/CSS, React, FastAPI

Libraries: pandas, NumPy, Matplotlib, Seaborn